



Course Guidebook for Engineering Mechanics II (Dynamics)

1. GENERAL INFORMATION OF THE COURSE

Course Name:	Engineering Mechanics II (Dynamics)
Course Code:	MEng 1062
Course Credits Points:	5 cp
Contact Hours:	2 hrs lecture & 3hrs laboratory per week
Course Module:	Engineering Mechanics
Course Category:	Common
Status of Course:	Compulsory
Pre-Requisites:	CEng 1061 Engineering Mechanics I (Statics); Applied Mathematics I
Students' Department:	Mechanical Engineering & Industrial Engineering
Degree Program:	BSc in Mechanical and Industrial Engineering
Students' Admission:	Regular
Year of Study:	2
Semester:	Second Semester
Academic Year:	2024/2025
Course Duration:	From March 2024 to June 2024
Course Chair:	Berihun Abebaw
Course Instructors:	Aragaw Mulu, MSc in Manufacturing Engineering Email: aragmulu@gmail.com Office: Agri: 007 Office: Agri: ____
Consultation:	Monday morning 10:00AM -11:00AM Thursday afternoon 2:00PM-3:30PM

2. COURSE DESCRIPTION

This course introduces students to the fundamental principles of dynamics, focusing on the motion of particles and rigid bodies. Topics include kinematics, kinetics, work-energy methods, impulse-momentum principles, and the application of these concepts to solve engineering problems. Upon completion, students will be able to analyze and solve problems related to the motion of objects under various forces and constraints.

3. OBJECTIVES OF THE COURSE

This course enables the students: In particular, the specific objectives the courses are:

1. Develop an understanding of the basic concepts of dynamics, including equations of motion and gravitation.
2. Analyze the kinematics of particles and rigid bodies using different coordinate systems.
3. Apply Newton's laws, work-energy principles, and impulse-momentum methods to solve engineering problems.
4. Understand the motion of rigid bodies, including rotation and relative motion.
5. Develop problem-solving skills through practical applications of dynamics principles.

4. COURSE LEARNING OUTCOMES (CLO)

At the end of the course, students will be able to:

N	Course Learning Outcome (CLO)	Performance	Program
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o		Indicators (PIs)	Student Outcome (PSO)
1	Analyze the motion of particles and rigid bodies using kinematic principles.	1.1, 1.2, 1.3, 2.1, 2.2	1,2,6
2	Apply Newton's laws to solve problems involving forces and motion.	2.1, 2.2, 3.1, 3.2	1, 2, 3
3	Use work-energy and impulse-momentum methods to analyze dynamic systems.	3.1, 3.2, 4.1, 4.2	1, 2, 4
4	Understand and analyze the motion of rigid bodies, including rotation and relative motion.	4.1, 4.2, 5.1, 5.2	1, 2, 5
5	Solve engineering problems involving dynamics principles.	5.1, 5.2, 6.1, 6.2	1, 2, 6

5. THE COURSE TIME SCHEDULE AND TEACHING METHODOLOGY

The following topics will be covered in this course within this semester.

Table 1. Time Schedule and Teaching Methodology of the course

Time Schedule	Course Contents	Teaching Methodology and References	Attaining student outcome
Week 1	1. Chapter 1: Introduction 1.1 - Basic concepts; 1.2 Equations of motion; 1.3 Gravitation.	- Lecture, - Tutorial; examples should be done. - Ref. 1 - Ref. 2	Outcome 1
Week 2-6	2. Chapter 2: Kinematics of Particles 2.1. Rectangular motion; 2.2. Plane curvilinear motion; 2.3. Coordinate systems; 2.4. Relative motion; 2.5. Constrained motion.	- Lecture, - Tutorial; examples should be done. - Ref. 1 - Ref. 2	Outcome 2
	Mid exam	Assessment	Outcome 1&2
Week 7-10	3. Chapter 3: Kinetics of Particles 3.1. Newton's second law; 3.2. Work-Energy equation; 3.3. Impulse and Momentum; Impact.	- Lecture, - Tutorial; examples should be done. - Ref. 1 - Ref. 2	Outcome 2, 3
Week 11-13	4. Chapter 4: Kinematics of Rigid Bodies 4.1. Fixed axis rotation; 4.2. Absolute motion; 4.3. Relative motion.	- Lecture, - Tutorial; examples should be done. - Ref. 1 - Ref. 2	Outcome 4
Week 14-15	5. Chapter 5: Kinetics of Rigid Bodies 5.1. General equations of motion; 5.2. Work-Energy method; 5.3. Impulse and Momentum.	- Lecture, - Tutorial; examples should be done. - Ref. 1 - Ref. 2	Outcome 4
Week 16	Final Exam		Outcomes 1-5

6. ASSESSMENT AND EVALUATION

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, site visit reports, homework, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines. If you encounter any difficulties with the assignments or tests, please contact the instructor promptly for assistance.

Home works, Assignments and project: Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete their own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur in penalties. All group members must contribute equally. Timely submission is mandatory, as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lecture or tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During exam all necessary materials will be provided by examiner.

Table 2. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1	Assignment 1	13	Chapter 1 -4	1	Week 1
2	Assignment 2	12	Chapter 2	3	4
4	Mid exam	25	Chapter 1-2	2,4	
13	Final Exam	50%	Chapter 3-5	1,2,3,4,5,	16
	Total	100%			

7. LABORATORY/WORKSHOP/ SESSION

8. COURSE POLICY

All students are expected to adhere to the code of conduct for students as per the Senate Legislation of Bahir Dar University, December 2020, during the duration of this course. Academic misconduct, which includes cheating, fabrication, and plagiarism, will not be tolerated and will be promptly reported to the relevant authorities for appropriate action.

The teaching methodology for this course will be diverse, encompassing activities such as reading, lectures, tutorials, construction site visit, discussions, home work, classwork exercises, and presentations, etc. Active participation from students is expected, including asking questions and raising issues whenever clarification is needed.

Regular attendance and punctuality are expected from all students. Those who miss more than 15% of the semester's classes will not be eligible to sit for the final exam. Attendance will be recorded for each class, and students are responsible for all announcements and changes made in class. Any student who exceeds the 15% absence threshold will receive a grade of 'NG' (No Grade) and must provide acceptable reasons for their absence, as per Senate legislation, Art. 174 for NG management.

Cell phones brought to class must be set to silent mode to avoid disruption to the teaching-learning process. Any form of disturbance during class time is unacceptable and must be avoided.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or complete a term paper/project without a valid reason, the instructor will record an "NG" according to senate legislation, Art. 171.

It is important to carefully follow the instructions provided in your course guidebook for completing all requirement of the course.

9. GRADING SYSTEM

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

Table 3. Grading scale of the course

Raw Mark Interval (100%)	Corresponding Letter Grade	Corresponding Fixed Number Grade
$90 \leq x \leq 100$	A ⁺	4.0
$85 \leq x < 90$	A	4.0
$80 \leq x < 85$	A ⁻	3.75
$75 \leq x < 80$	B ⁺	3.50
$70 \leq x < 75$	B	3.00
$65 \leq x < 70$	B ⁻	2.75
$60 \leq x < 65$	C ⁺	2.50
$50 \leq x < 60$	C	2.00
$45 \leq x < 50$	C ⁻	1.75
$40 \leq x < 45$	D	1.00
$x < 40$	F	0.00

10. TEXTBOOKS

Meriam J.L., Engineering Mechanics - Dynamics, 6th ed., 2003.

11. REFERENCES

1. Hibbeler, Rusel M., Engineering Mechanics: Dynamics, 10th ed., 2003.
2. Beer, Johnston, Clausen, Eisenberg, Cornwell, Vector Mechanics for Engineers: Dynamics, 9th ed., 2004.

12. APPROVAL

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Aragaw Mulu		
Course Module Chair			
Chair Holder (Manufacturing Engineering)	Berihun Abebe		
V/Dean			

Notes:

Student outcomes and course learning outcomes are related but distinct concepts in education:

1. **Student Outcomes:** These are the specific knowledge, skills, or attributes that students are expected to demonstrate or achieve by the end of a program or course. They focus on what students will be able to do or demonstrate as a result of their learning experiences. Student outcomes are often aligned with broader educational goals and may include abilities such as critical thinking, problem-solving, effective communication, and subject-specific skills.
2. **Course Learning Outcomes:** These are the specific statements that describe what students should be able to know or do as a result of completing a particular course. Course learning outcomes are more specific and focused on the content and objectives of the course itself. They outline the expected achievements or competencies that students should attain by the end of the course. These outcomes guide course design, instruction, assessment, and evaluation.

In summary, student outcomes are broader and pertain to the overall goals of an educational program, while course learning outcomes are more specific and relate directly to the content and objectives of individual courses within that program.

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In education curriculum design, both student outcome performance indicators (PIs) and course learning outcomes (CLOs) are important components, but they serve slightly different purposes:

1. **Student Outcome Performance Indicators (PIs):**
 - PIs are specific, measurable criteria used to assess students' attainment of learning outcomes.
 - They provide detailed descriptions of what students should be able to demonstrate or accomplish to meet the learning outcomes.
 - PIs often include observable behaviors, skills, or knowledge that can be assessed through various methods such as exams, assignments, projects, presentations, or observations.
 - PIs help educators and curriculum designers develop assessment strategies and tools to evaluate students' progress and achievement.
 - PIs are often aligned with course objectives and standards to ensure that assessment activities effectively measure student learning.
2. **Course Learning Outcomes (CLOs):**
 - CLOs are broad statements that describe what students are expected to learn or achieve by the end of a course or program.
 - They represent the overarching goals and purposes of the course and provide a framework for instructional planning and assessment.
 - CLOs articulate the knowledge, skills, abilities, or attitudes that students should develop as a result of their learning experiences.
 - CLOs are typically derived from broader educational goals, program objectives, or professional standards.

- CLOs guide the selection of instructional content, activities, and assessments to ensure that they are aligned with desired learning outcomes.

In summary, PIs are the specific criteria used to measure student achievement of CLOs. While CLOs provide a high-level overview of desired learning outcomes for a course or program, PIs offer more detailed, observable indicators of student performance that can be used to assess progress and attainment of those outcomes.